ASSIGNMENT FOR SENIOR PROJECT STUDENTS

Courses: Senior Project Series (EGR 4810, EGR 4820, EGR 4830, and Dept. Specific Course (e.g. CHE 4631)

Learning Skills Assessed: Critical Thinking, Communication Skills, Information Literacy and Quantitative Literacy

Purpose: The objective of this assignment is to assess engineering students' critical thinking, written communication, information literacy skills, and quantitative literacy. The individual student's assessment data is confidential; this is accomplished by redaction of all identifications prior to evaluation. The evaluation is done outside of CANVAS to ensure confidentiality.

Learning Objectives:

This assignment will assess engineering students' achievement of the following GE- Student Learning Outcomes and ABET- Student Outcomes:

GE -SLO	Engineering Program Student Learning Outcomes (ABET SOs)
Written Communication	**SLO 3** - an ability to communicate effectively with a range of audiences.
Critical Thinking	**SLO 6** - an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
Quantitative Reasoning	**SLO 1** - an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics **SLO 2** - an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
Information Literacy	**SLO 7** - an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Prompt: You are considered for an on-site job interview and are required to discuss your senior project work. The panel is skeptical about your findings and has requested that you write a technical memo (or executive summary) to explain the motivation of the project, the rational for the proposed solution and how the findings meet the needs of the client/community.

Task: Each student enrolled and completing the senior project series (EGR 4810,4820 and 4830) shall submit **an executive summary** of their project that addresses the listed topics below in the following order:

1. Title page containing the following (use template provided in Canvas):
    a. Title of your Project - Executive Summary
    b. Your full name
    c. Bronco ID
    d. Your Faculty Advisor's Name
    e. Your degree program
    f. Date of Submission
    g. How long it took you to write this executive summary
    h. Did you attend or watch the lesson from Mr. Paul Hottinger? (YES/NO)

2. Define the **objective** of the project by identifying the problem(s)/issue(s)/need(s).
3. Student identifies and analyzes other perspectives (arguments) and presents his/her own perspective or position (**i.e. literature review**).
4. Student to outline the methodology used to develop the solution.
5. Students use(s) quantitative information, evidence and data to support their position(s), which address the problem/question through discussion of their results. Through this process there should be justification and validation of results being presented. (This can include data sources, figure/data plot or table that assists your ability to discuss your final results or final design. Clearly label all figures/data plot or table with a number and descriptive title. All Figures and Tables should be included in Appendix A and reference in the text. Figure numbers and titles go below the figure, table numbers and titles go above the table).
6. Student draws adequate conclusion(s) given their result(s) (This could include pros and cons of your final design, application of design and/or recommendations for future design. Which is supported by numerical values (i.e results) as evidence to support your conclusion.)
7. **Work** Cited (select a specific referencing convention: IEEE, AICHE, AIAA, ASME, ASCE, IISE, SME, and ASEE). Remember to include in-text citation and it should be connected to your work cited page.
8. Appendix A – Tables and Figures
9. Appendix B – One Example of a Sample Calculation (remember to reference the sample calculation in the text of the executive summary)

Instructions for the write-up

- Your write-up should be structured in a series of coherent paragraphs. You may or may not use section headers. If you would like to use section headings, please use appropriate terminology such as: **Introduction** *(purpose, objective, and scope of the problem or issue identified)*, **Literature Review** *(present your and other perspectives)*, **Methodology**, **Results and Discussion** *(provides data, analysis of data and discussion of results)*, and **Conclusion**. Remember to add work cited and sample calculations.
- Your write-up should be approximately 1,250 to 2,000 words. The following are additional items required in the submittal, which do not count towards the word count: Work Cited, Appendix A and Appendix B.
- Your write-up should be single-spaced, Times New Roman, 12-point Font and formatted to 1-inch margins (use template provided).
- You will submit this assignment electronically to your instructor and on Canvas.
- The assignment deadline is provided on the Canvas website.

THIS ASSIGNMENT WILL BE A CERTAIN PERCENTAGE (left to the discretion of your project advisor) OF YOUR SENIOR PROJECT GRADE